

Suppose that we have $v_1, \dots, v_m$ in R^n . How can we tell if they span all of R^n ?

Answer: If we form $A = (v_1 \ v_2 \ \dots \ v_m)$ then $v_1, \dots, v_m$ spans R^n iff when you row reduce the matrix A , every row has a leading 1.

Important fact: If $v_1, \dots, v_m$ spans R^n then $m \geq n$.

Properties of Multiplication of Matrices and Vectors

Theorem

Suppose that A is $n \times m$ and x, y are in R^m .

- 1 $A(x + y) = Ax + Ay$ and,
- 2 for every number λ , $A(\lambda x) = \lambda Ax$.

Homogeneous systems

- We say that a system of linear equations is a homogeneous system if all the constants in the equations (the right-hand sides) are 0. A homogeneous system can be written as $Ax = 0$.
- Any homogeneous system of linear equations has at least one solution; that is, $x = 0$.
- Theorem: Any homogeneous system with more unknowns than equations has infinitely many solutions.

Solutions to linear systems using homogeneous equations

Theorem

Suppose that $Ax = b$ is some consistent system of linear equations and x_0 is some solution. Then every solution to $Ax = b$ has the form $x_0 + y$ where y is a solution to the homogeneous equation $Ax = 0$.